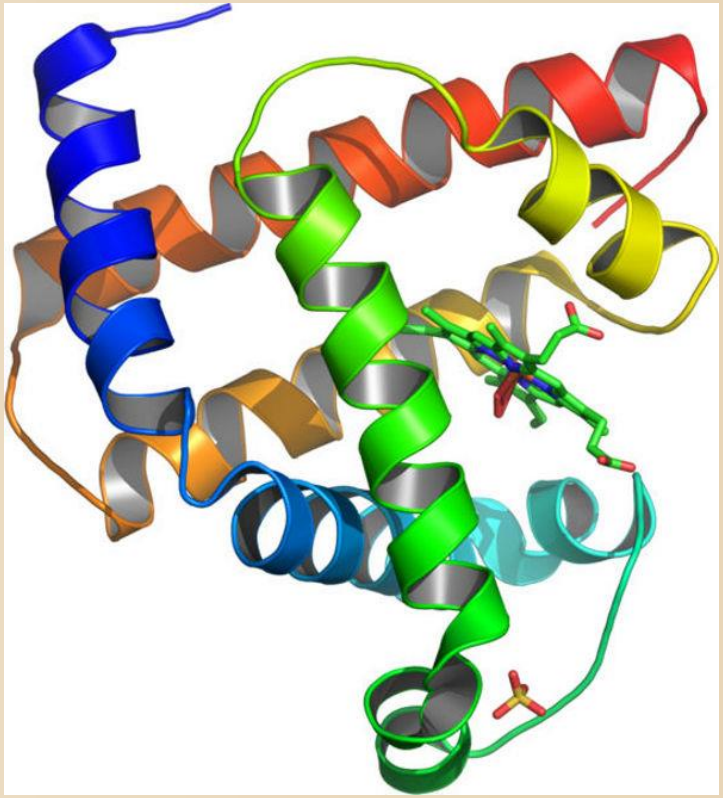
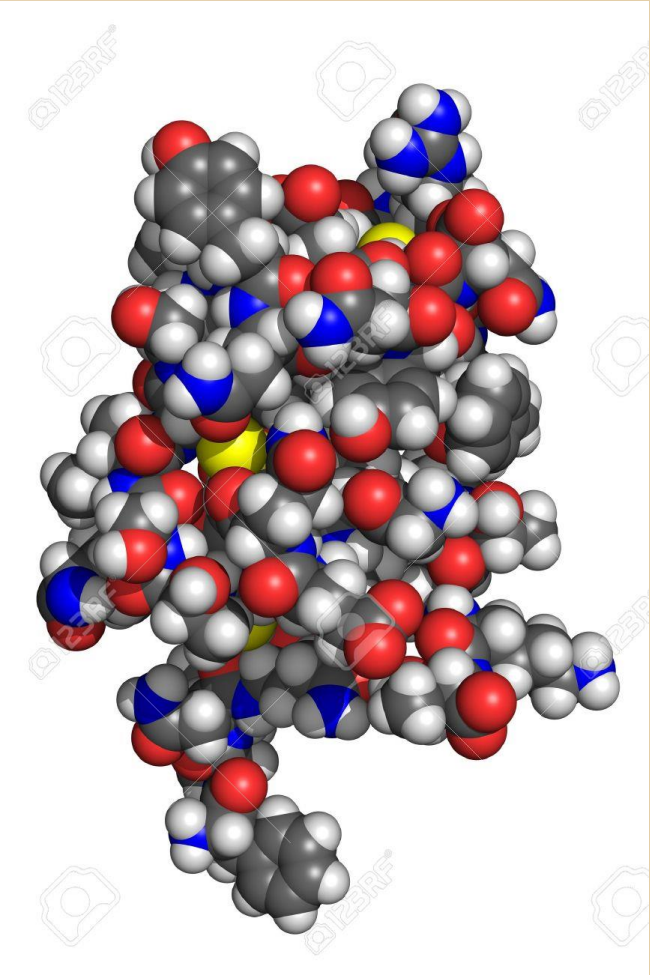
Organic Chemistry
1800 – 1900





Before

We talked about Chemistry of Non-Living

Before 1800, only a small number of organic compounds were known such as waxes, fats, oils, acetone, sugars, oxalic acid, urea, and alcohol.

According to the vitalist theory, organic compounds had to be found in nature, they could not be made in the laboratory.

Chemists that did this kind of work became known as natural product chemists.

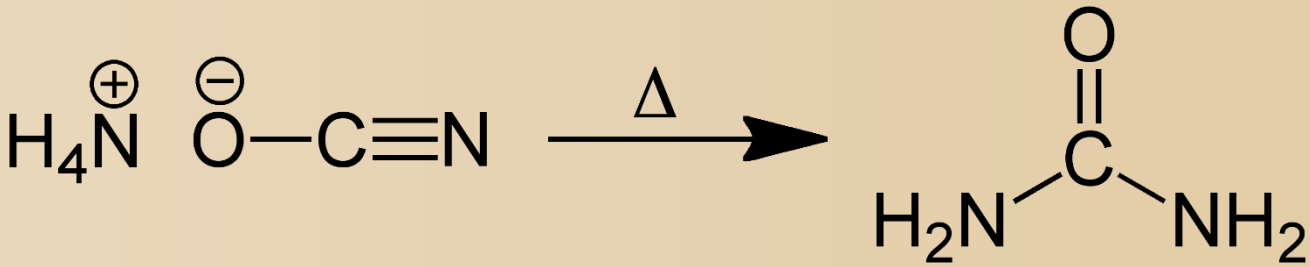
Discovery of Urea

Urea is found in mammalian urine. (It is produced by mammals to remove excess nitrogen.)

Friedrich Wöhler (1828) was a young German chemist who set out to make organic compounds in the laboratory. He wished to disprove the Vitalist theory.

Wöhler wrote "...I can prepare urea without requiring a kidney of an animal, either man or dog."

Synthesis of Urea



Organic Chemicals

Lavoisier made the first systematic studies of organic composition in 1786.

He defined organic chemicals as combinations of **oxygen** with radicals that contained **carbon** and **hydrogen**.

If of animal origin, they could also contain nitrogen and phosphorus.

Lavoisier determined carbon content by combustion.

Organic Chemistry

Gay-Lussac and Thenard found that the ratio of hydrogen to oxygen in sugars, starches, woods, and gums was the same as in water, namely two hydrogen atoms for each oxygen atom. However, these compounds did not contain water units

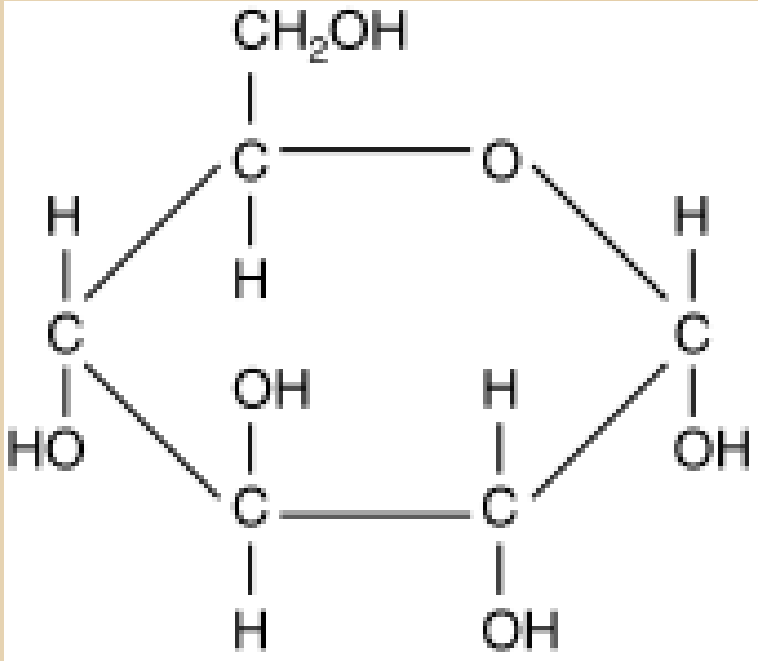
- Different arrangements of the atoms to have different structures

Berzelius analyzed a sufficient number of organic compounds to prove that they obeyed the same laws as inorganic compounds.

Carbonhydrates

In addition to hydrogen and oxygen compounds like sugars, starches, woods contain carbon

Hence, they acquired the name carbonhydrates.



Chemical Characterization

At that time, chemists characterized chemical compounds by elemental analysis, color, crystal form, melting point, boiling point, taste, smell, and elemental analysis.

When ammonium cyanate is heated, it rearranges to urea that has a melting point of 132.7 C. While the elemental analysis gives same formula (CH_4 ON_2), the properties of the two compounds are quite different.

Structure

A theory for the structures of organic molecules was needed. Dumas and others proposed various radicals such as etherin (ethylene, $-C_2H_4-$) that could have things added to them to become compounds such as ethyl alcohol, ethyl ether, ethyl acetate, ethyl chloride, and so on.

While the etherin theory was not complete, it came very close by proposing such correct reactions as:

- C_2H_4 (ethylene) + H_2O (water) = C_2H_6O (ethyl alcohol or ethanol). In fact, this very reaction is used today in an industrial process for making ethanol.

Stereochemistry

In 1837, Auguste Laurent developed the nucleus theory in which there was a three-dimensional structure in the form of radical cubes with atoms added to or removed from the faces to give different properties.

This allowed such exchanges as chlorine replacing hydrogen without a change in the geometry.

It was also the beginning of modern stereochemistry (3D Structure).

Stereochemistry

Laurent had actually been the first, in 1846, to propose that elementary gases such as oxygen and nitrogen were diatomic molecules.

Diatomic molecules are molecules composed of only two atoms, of the same or different chemical elements.

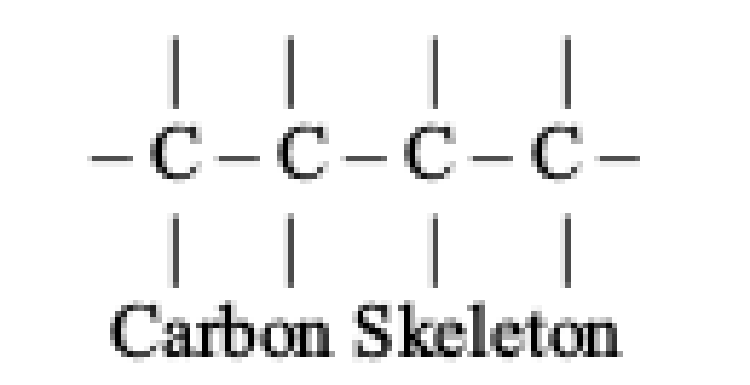
If a diatomic molecule consists of two atoms of the same element, such as hydrogen (H_2) or oxygen (O_2), then it is said to be homonuclear.

Carbon Skeleton

Carbon could have four bonds (Valence).

Friedrich Kekulé (1858) dreamed of carbons forming a chain by connecting to itself or to other atoms to make organic molecules.

This carbon skeleton with various function groups attached gave an explanation for the variety of properties seen in organic compounds.



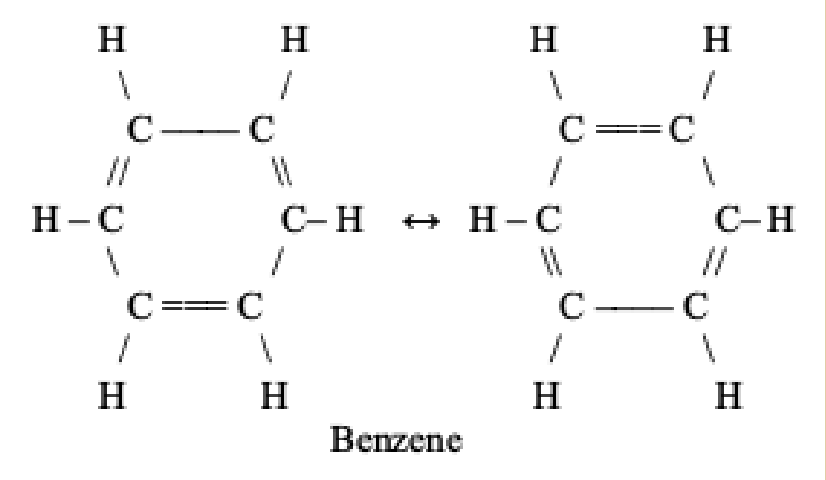
Benzene & Aromatic Compounds

A few years later Kekulé would dream again, this time about snakes eating each other's tails, and gain insight into the structure of the unusual molecule benzene.

Benzene gives rise to an important class of compounds that are called aromatic molecules.

Due to distinctive smells

- they are called aromatic



Crystals

Pasteur observed that Certain compounds rotate polarized light.

Polarized light was either rotated to the right or to the left due to the arrangement of atoms in the Crystal.

If compounds don’t rotate polarized light, then they were equal mixtures of the two forms

Chemical Revolution in a Nutshell

Following Dalton, the empirical and theoretical thinking brought logical models which would be the mainstay of chemical understanding.

Wöhler, in a purposeful search to synthesize a natural product succeeded and the Vitalist theory that separated organic from inorganic chemistry is explained.

Theories of bonding to explain and predict chemical compounds and properties, were sought.

Chemical Industry – Dupont (1804)

A chemical company was established in Newark, Delaware, by Pierre du Pont de Nemours.

Du Pont had emigrated to America with his sons to avoid the fate of Lavoisier. Du Pont opened a chemical factory initially making gun powder to sell to Napoleon.

One important aspect of the rapid growth of the chemical industry was the research that was fostered by the commercial success of chemistry.

Besides medicine, chemistry was the first area of science to become a major industry and to attract industrial funding.

The first patent issued by the U.S. Patent Office was in chemistry.

Industrial Age

Because of the foundations of the Scientific Revolution and the Enlightenment, important discoveries were made in all areas of science.

During the industrial age, not only was science being used to make material progress but scientific discovery was happening at an unprecedented rate.

The chemical industry, the 19 th century was to know hundreds of thousands of new compounds synthesized in the laboratory by organic chemists. Aspirin was patented by George Bayer

Modern universities were founded that joined in research activities and the industrial world realized the economic advantages of research.

The Mystery of Matter

INTO THE ATOM

Learn how Harry Moseley's investigation of atomic number redefined the Periodic Table, and

Learn how Glenn Seaborg's discovery of plutonium opened up a whole new realm of elements still being explored today.